Deep Learning Workshops

Prof Thomas Lampert
lampert@unistra.fr

Dr Mihailo Obrenović
mobrenovic@unistra.fr

Télécom Physique Strasbourg
Université de Strasbourg

Université de Strasbourg

École d'ingénieurs Télécom Physique Strasbourg

iCUBE

Course Organisation

- 20h CI
- Assessment
  - Handin (project code) – 20%
  - Presentation (15 minutes + 5 mins questions) – 14%

Content

- Challenges
- Data Science
- Algorithm Evaluation

Summary of topics

Computer vision (CV)	6 projects
Natural Language Processing (NLP)	3 projects
Audio Signal Processing	2 project
Time series	1 project

Project List 1/2

Natural Language Inference	NLP	Inference, Classification
Emoji prediction	NLP	Classification
Fake News Detection	NLP	Binary classification
Ships	CV	Satellite images, Binary classification
Breast cancer classification	CV	Histology images, Binary classification
Sex and Age Prediction	CV	Binary and interval classification

Project List 2/2

Driver Drowsiness Detection	CV	Binary classification
Traffic Sign Recognition	CV	Classification
Urban street scenes understanding	CV	Semantic segmentation
Speech Emotion Recognition	Audio	Classification
Music Genre Classification	Audio	Classification
Stock Prediction	Time series	Forecasting

Natural Language Inference (NLI)

- Pairs of sentences given
- Task: What is the relation between them?
  - 0 - Entailment (implication)
  - 1 - Neutral (unrelated)
  - 2 - Contradiction
- Dataset:
  - 12K+ pairs of sentences

	A	B	C	D	E	F
1	**id**	**premise**	**hypothesis**	**lang_abv**	**language**	**label**
2	5130fd2cb5	and these comments were considered in formulating the interim rules.	The rules developed in the interim were put together with these comments in mind.	en	English	0
3	5b72532a0b	These are issues that we wrestle with in practice groups of law firms, she said.	Practice groups are not permitted to work on these issues.	en	English	2
4	3931fbe82a	Des petites choses comme celles-là font une différence énorme dans ce que j'essaye de faire.	J'essayais d'accomplir quelque chose.	fr	French	0

Emoji prediction

- Text messages given
- Task: Predict an emoji that suits the given text
- Dataset:
  - 132 sentences, 5 possible emojis

	A	B
1	never talk to me again	😞
2	I am proud of your achievements	😄
3	It is the worst day in my life	😞
4	Miss you so much	❤
5	food is life	🍴

Fake News Detection

- Given article and author
- Task: Detect if the article is:
  - 0 – reliable
  - 1 – unreliable
- Dataset:
  - 20K articles

Benoît Hamon Wins French Socialist Party's Presidential Nomination - The New York Times

Alissa J. Rubin

PARIS — France chose an idealistic, traditional candidate in Sunday's primary to represent the Socialist and parties in the presidential election this spring. The candidate, Benoît Hamon, 49, who ran on the slogan that he would "make France's heart beat," bested Manuel Valls, the former prime minister, whose campaign has promoted more policies and who has a strong background…

reliable

Russian Researchers Discover Secret Nazi Military Base 'Treasure Hunter' in the Arctic [Photos]

Amando Flavio

The mystery surrounding The Third Reich and Nazi Germany is still a subject of debate between many observers. Some believe that Nazi Germany, under the control of Adolf Hitler, possessed supernatural powers, and largely employed pseudo-science during the 1933-1945 period. However, some also hold that the above belief is just a mere speculation without any proven fact. Over the years, researchers have searched extensively for answers to some of the more mysterious activities associated with Nazi Germany…

unreliable

Ships

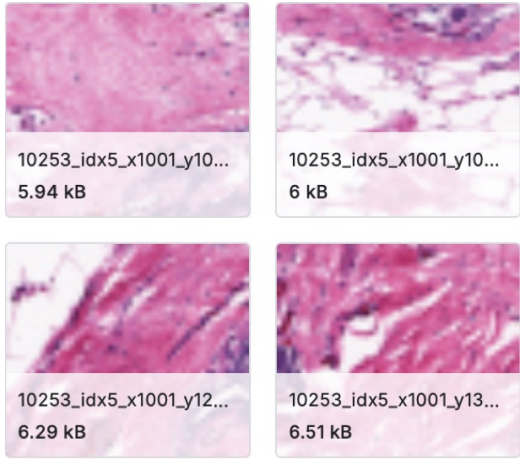
- Satellite images over bays
- Task: Detect ships
- Dataset:
  - 4K images (1K ship, 3k no-ship)
  - Size 80x80

ship

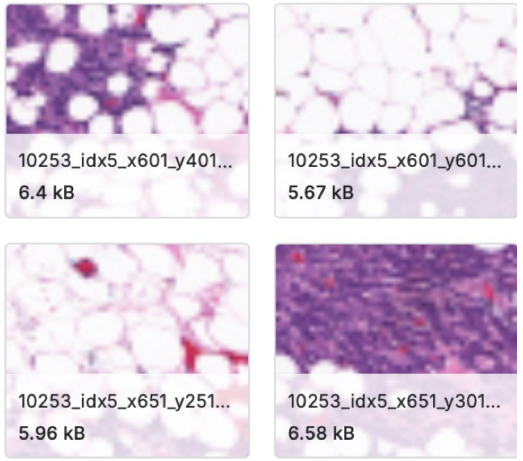
no-ship

Breast cancer classification

- Histopathological images
- Task: Classify if image is:
  - Benign
  - Malignant
- Dataset:
  - 270K patches (cut from 162 whole slide images)
  - Patch size 50x50



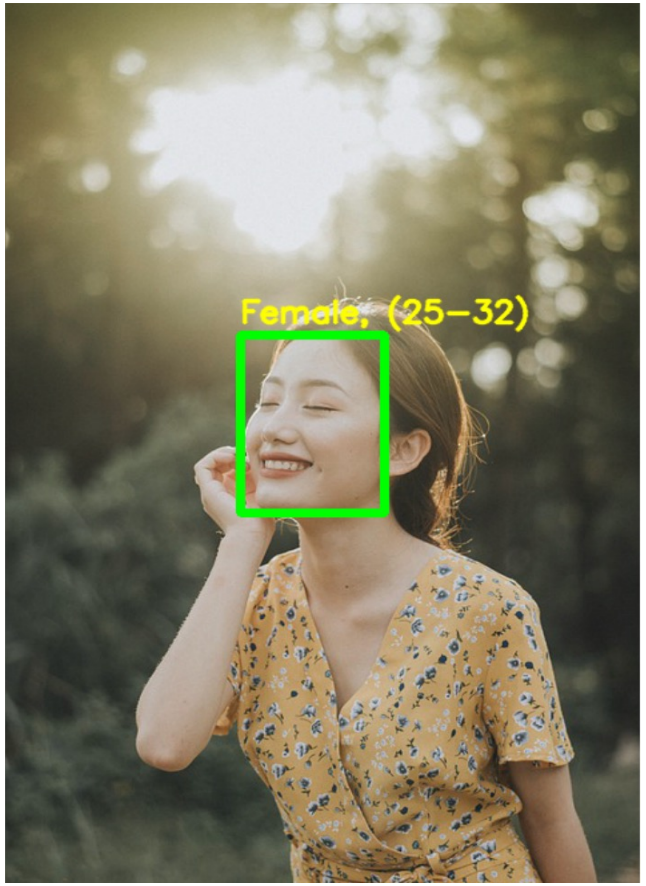
benign



malignant

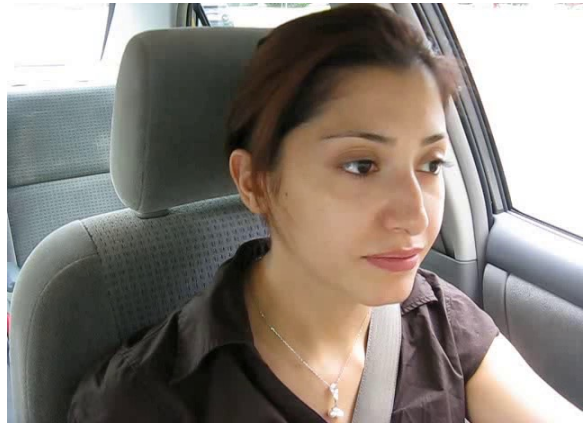
Sex and Age Prediction

- Images of persons
- Task 1: Predict if a person is male or female
- Task 2: Predict the age of a person
  - 8 predefined intervals (0-2), (4-6), … , (60-100)
- Dataset:
  - 26K face photos
  - Varying lighting, pose, appearance, …



Driver Drowsiness Detection

- Detection system to alert the driver when he feels sleepy
- Dataset 1 – driver images
  - Task - detect if the driver is yawning or not
- Dataset 2 – eyes images
  - Task – Detect if the eye is open or closed
- ~600 images per class

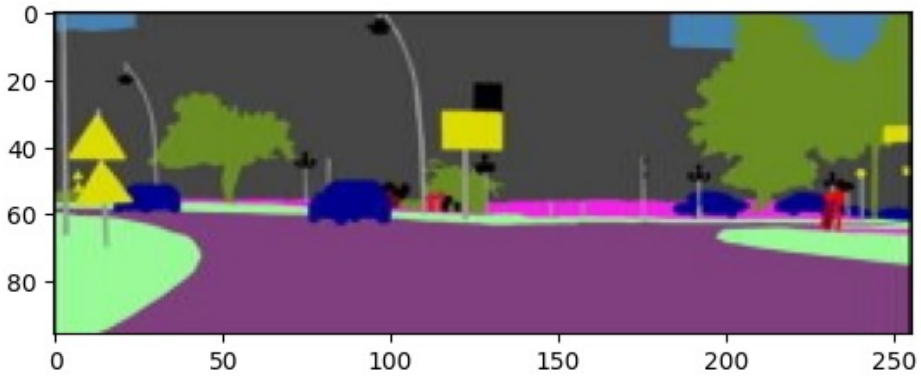


Traffic Sign Recognition

- Self-driving cars need to understand traffic signs
- Dataset – traffic sign images
  - 43 different classes
  - 50K images
  - Imbalanced dataset
- Task – Recognize the traffic sign

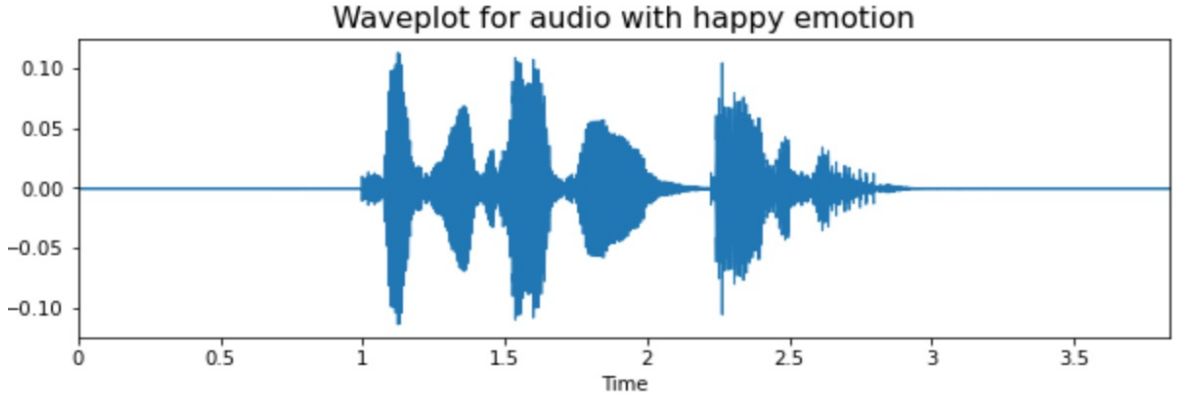
Urban street scenes understanding

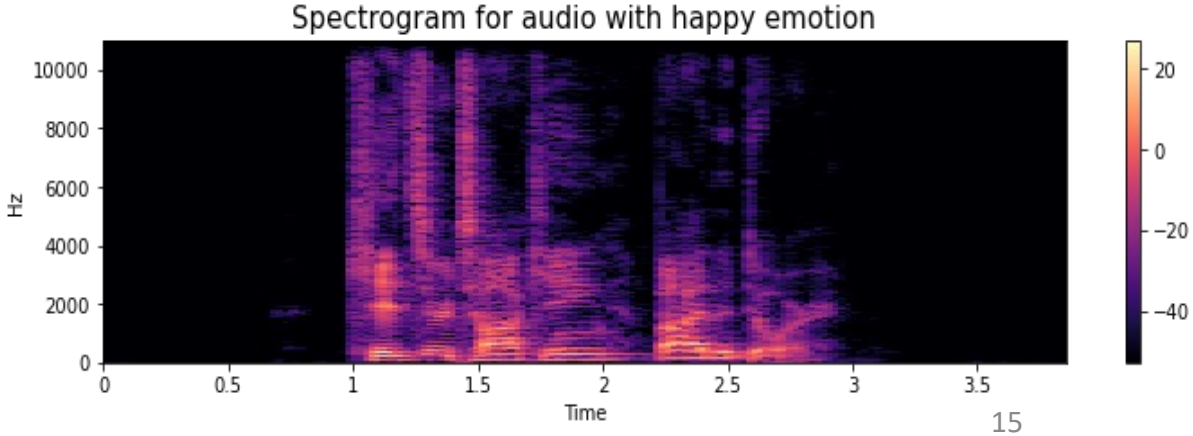
- Semantic understanding of urban street scenes
- Dataset - images taken from vehicles + their segmentation masks
  - 3500 images and masks
  - 30 classes
- Task – assign labels to each pixel of the image (semantic segmentation)



Speech Emotion Recognition

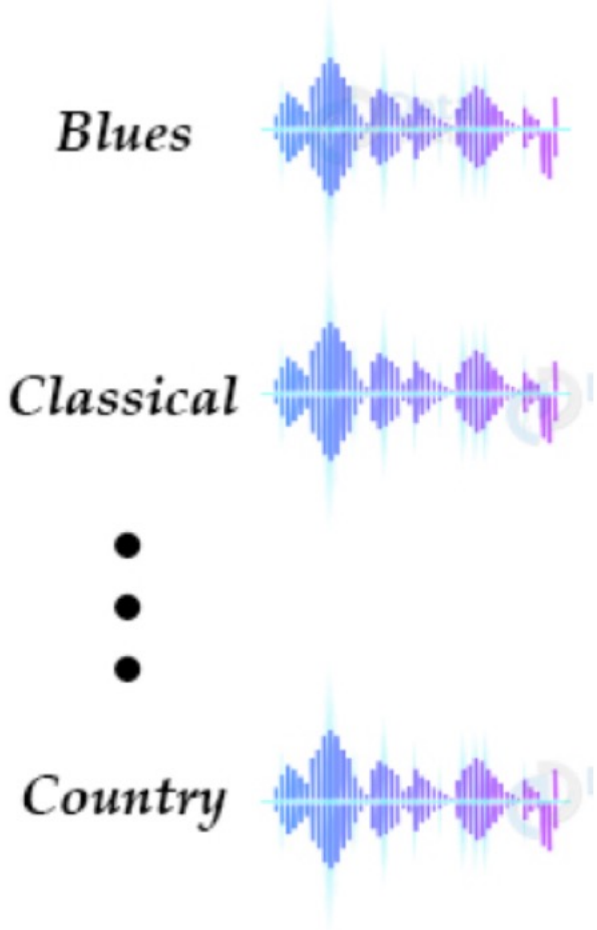
- Recognize human emotion from speech
- Dataset – 7K clips
  - 91 actors
  - Both male and female, a variety of races and ethnicities
  - Given audio files and Mel spectrograms
- Task – Recognize one of the six emotions
  - Anger, Disgust, Fear, Happy, Neutral, and Sad





Music Genre Classification

- Classify different musical genres from audio files
- Dataset – 1000 audio music tracks
  - 30 seconds duration
  - 10 classes, 100 tracks each
  - .wav files and Mel spectrograms
- Task – Recognize one of 10 genres
  - Blues, Classical, Country, Disco, Hiphop, Jazz, Metal, Pop, Reggae, Rock



Stock Prediction

- Predict future stock values of big companies
- Dataset - historical stock market data
  - 5 years (2018-2023)
  - ~500 top companies
- Task – based on the stock values of a company during the previous N days, predict the stock value for the day N+1.

	A	B	C	D	E	F	G	H	I
1	**Date**	**Open**	**High**	**Low**	**Close**	**Volume**	**Dividends**	**Stock Splits**	**Company**
2	2018-11-29 00:00:00-05:00	43.829760572993	43.8633538041636	42.6395935832266	43.0835075378418	167080000	0	0	AAPL
3	2018-11-29 00:00:00-05:00	104.769074332185	105.519257086357	103.534594914971	104.636131286621	28123200	0	0	MSFT
4	2018-11-29 00:00:00-05:00	54.1764984130859	55.0074996948242	54.0999984741211	54.7290000915527	31004000	0	0	GOOGL
5	2018-11-29 00:00:00-05:00	83.7494964599609	84.4994964599609	82.6165008544922	83.6784973144531	132264000	0	0	AMZN
6	2018-11-29 00:00:00-05:00	39.6927840259795	40.0649038762231	38.7351954599368	39.0378532409668	54917200	0.04	0	NVDA

Choose your challenge

https://bit.ly/4iGNd1I

- Teams of 3

Python Packages

- General – Scikit-learn, tensorflow, keras

- Image – Scikit-image
- Time-Series – Sktime
- NLP – NLTK, TextBlob, Pattern